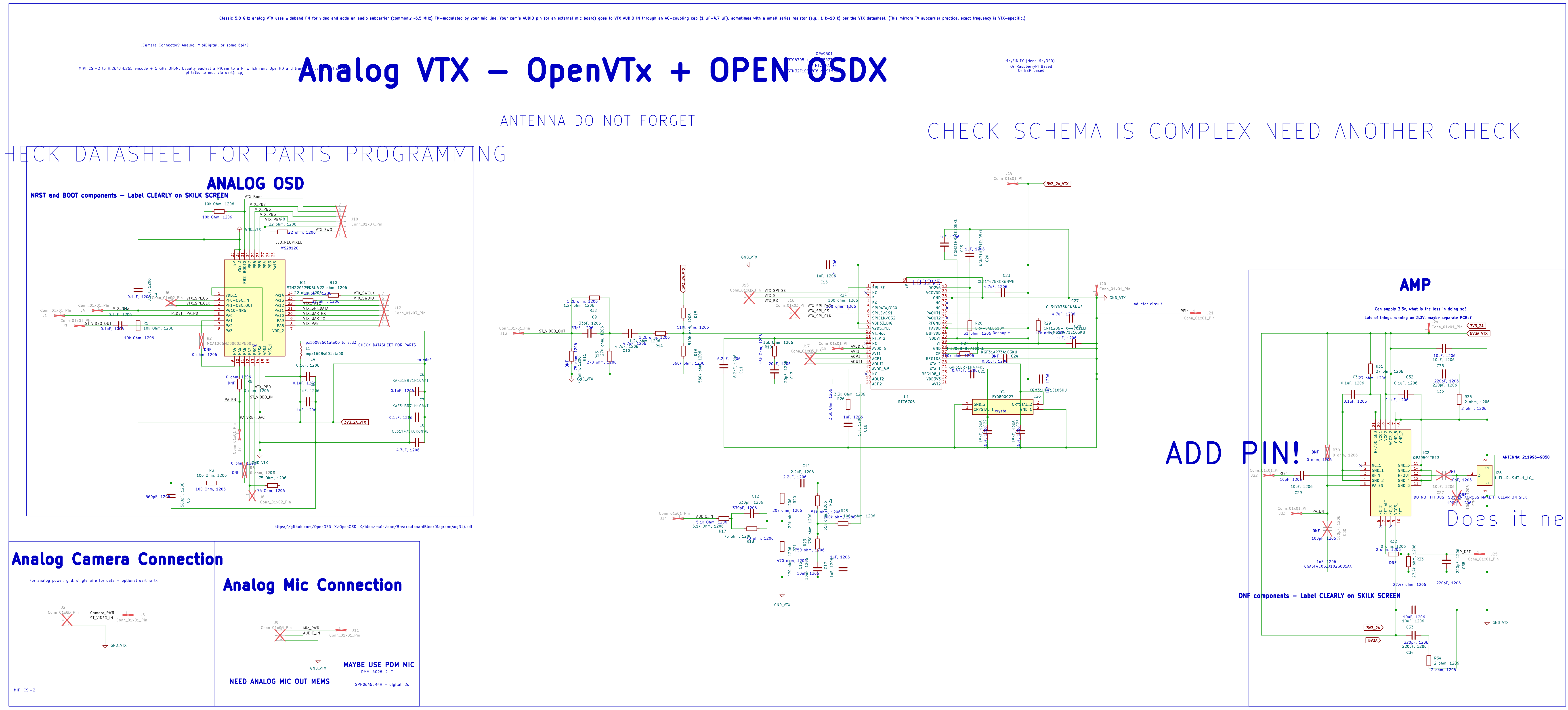
Analog VTX - OpenVTx + OPEN OSDX
ANALOG OSD
AMP
Analog Camera Connection
Analog Mic Connection
NRST and BOOT components - Label CLEARLY on SKILK SCREEN
DNF components - Label CLEARLY on SKILK SCREEN
MAYBE USE PDM MIC
DMM-4026-2-T
SPH0645LM4H - digital i2s
NEED ANALOG MIC OUT MEMS
MIPI CSI-2
QPA9501
tinyFINITY (Need tinyOSD)
Or RaspberryPi Based
Or ESP based
DO NOT FIT JUST SOLDER ACROSS MAKE IT CLEAR ON SILK

Classic 5.8 GHz analog VTX uses wideband FM for video and adds an audio subcarrier (commonly ~6.5 MHz) FM-modulated by your mic line. Your cam's AUDIO pin (or an external mic board) goes to VTX AUDIO IN through an AC-coupling cap (1 µF-4.7 µF), sometimes with a small series resistor (e.g., 1 k-10 k) per the VTX datasheet. (This mirrors TV subcarrier practice; exact frequency is VTX-specific.)

.Camera Connector? Analog, MipiDigital, or some 6pin?

MIPI CSI-2 to H.264/H.265 encode + 5 GHz OFDM. Usually easiest a PiCam to a Pi which runs OpenHD and transmits using Wi-Fi cards
pi talks to mcu via uart(msp)

Analog VTX - OpenVTx + OPEN OSDX

ANTENNA DO NOT FORGET

CHECK SCHEMA IS COMPLEX NEED ANOTHER CHECK

CHECK DATASHEET FOR PARTS PROGRAMMING

Can supply 3.3v, what is the loss in doing so?

Lots of things running on 3.3V, maybe separate PCBs?

ADD PIN!

Does it need RF filter?

For analog power, gnd, single wire for data + optional uart rx tx

https://github.com/OpenOSD-X/OpenOSD-X/blob/main/doc/BreakoutboardBlockDiagram(Aug31).pdf

360 cam dont need gimbal

For next edition place decouple caps close to pins properly

NextGen Board Size:
https://betaflight.com/docs/development/manufacturer/manufacturer-design-guidelines?utm_source=chatgpt.com

ATMEGA 328 p?
BMP280
8S ESC / 60A - FETs + Gate Drivers?

- UARTs: plan for RX, VTX control, GPS, DJI/HD, Bluetooth; expose at least 5.
- May Need DEDICATED BLE Module not a STM32WB5MMG

CHECK ALL COMPONENTS FOR BEST PCB DESIGN
May need to be 6 Layer for big power planes + 2 gnds and plenty of thermal vias!
Add fiducials for assembly orientation on silk screen

https://www.reddit.com/r/PrintedCircuitBoard/comments/1aqmsye/review_request_schematic_of_stm32f7_based_flight/

ToDo Pi & Extra:
DIGITAL VTX

STM Pogo Pin Programmer?

Electronic Switch + BTNs?

HDMI to analogue video + audio
MIPI Cam & Display + Usual Pi Periphs
CM5 Pi Connector?
SD Card Black Box as well as Pi OS (2x)
BMS - on board charger? may be not needed and extra cost
Make up to X8 + Fixed wing confix from these boards
Little Cap On Power Bus, Reverse Polarity on Battery, Fuses, More Filtering

Change AT7456E, RTC6705 + RTC6715 for something western

ToDo ESC:

Current Busbar
Filter Caps
Heatsinks

ToDo FC:

3-5 UARTS + SBUS (Invert UART) + SPI + I2C + USB
GPIOs, PCM PWM, ADC + DAC, CLK DIN DOUT, SWD?
Standard GPIO Pins/Connectors
CANBus? I2S? for flying music to peoples windows lol or a separate load?
RSSI Input - For Remote?
Controller RX e.g. FRSKY OSD Mini
4 Camera Pin? + 3 Motor Pin? Connectors

Current Busbar
Filter Caps
Heatsinks
4-6 Layer Board 2 OZ Copper?

SD Card

NB! Max used depends on - Motor Rating (e.g. 2-4S), ESC rating (mosfet) e.g. 28V = 6S, 60V = 8s + protection circuit max rating
XT90 = 8s bat = 33.6V = 90A +
XT60 = 6s bat = 25.2V = 60A
XT30 = 4s bat = 18V

PCIE Gen 2.0 - NVME

USE BETAFLIGHT CONNECTORS

CHECK DATASHEET FOR PARTS FOOTPRINTS WHEN DOING PCB

Sheet: /
File: Part2.kicad_sch
Title:
Size: A0
Date:
Rev:
KiCad E.D.A. 8.0.6
Id: 1/1